



EDUCATION			
Program	Institute	% / CGPA	Year
B. Tech in Chemical Engineering	Indian Institute of Technology Madras	8.99	2026
XII (Central Board of Secondary Education)	Delhi Public School South, Bangalore	98.2%	2022
X (Central Board of Secondary Education)	National Hill View Public School, Blore	96.4%	2020
Scholastic Achievements	- Achieved top 2 percentile among 2,00,000 aspirants in JEE Advanced 2022 showing noteworthy scholastic success. Proof - Ranked in the top 0.9 percentile in JEE Mains 2022 and top 0.4 percentile in Chemistry among 10,00,000 students. Proof - Secured 73th rank in ComedK, 84th rank in KCET and rank 35th in PESSAT amongst a pool of over 2,00,000 aspirants. Proof1 Proof2 Proof3 - Selected as an NTSE Scholar in 2020 and a KVPY Fellow in 2022 (SX) and secured National Rank 1 in UCO 2019. Proof1 Proof2 Proof3		
COURSES AND CERTIFIED SKILLS			
Courses	Supervised Machine Learning Proof	Scientific Computing	Artificial Intelligence
	Digital Video Processing	Probability	Machine Learning Techniques
	Multivariate Data Analysis	Computational Techniques	Linear Algebra
Courses	Process and Product Design	Applications of Mass Transfer	Process Dynamics and Control
	ML in Reaction Engineering	Material Science	Chemical Reaction Engineering
Skills	DNN, CNN, Data Preprocessing, Feature Engineering, Simulation, RAG	Python, C++, MATLAB, Shell Scripting, C#	OpenCV, Langchain, Scikit-learn, TensorFlow, Pandas, NumPy, Git
PROFESSIONAL EXPERIENCE			
Operations Intern Danone May '25 - Present (Extended Offer & Awarded PPI)	Sole member to receive a 3-month internship extension & PPI, collaborating with Danone's team across the globe. - Improved shopfloor query resolution time by 60% by developing an offline LLM chatbot (LoRA) trained on SOPs. - Prototyped an RAG framework with vector databases for scalable LLM integration into Danone's knowledge systems. - Reduced false rejection rates in 3 packaging lines by 80% through cameras, lighting & carton design improvements. - Increased analysis speed by 40% & eliminated manual errors by automating QC report generation using 3 tools. Proof1 Proof2		
AI/ML Intern Ravvio Jan '25 - May '25	Sole Junior in Gen AI chatbot development, AWS deployment, and multilingual transcription for enterprise clients. - Built multiple AI chatbot workflows in Python and Dialogflow, raising lead qualification accuracy by 18% in 3 pilots. - Deployed and optimized chatbot services on AWS EC2/S3, handling 25k+ monthly interactions with 99.9% uptime. - Processed transcribed 1,000+ calls using IndicWhisper, improving STT accuracy for 5+ regional languages by 15%. - QA-tested 500+ AI post-call summaries using Python scripts, fixing 15+ bugs and improving accuracy by over 20%. Proof		
AI Intern EXL Jun '24 - Oct '24	Sole Sophomore involved in the project Gen AI & Industrial Technology Integration for Enhanced Plant Maintenance - Researched AI and GenAI integration in industrial technology, aiming for a 30+% reduction in downtime and costs. - Developed predictive maintenance models with a projected accuracy of over 90% in forecasting equipment failures. - Created AI-driven maintenance protocols that's projected to enhance successful task completion rates by over 25%. - Conducted 10+ simulations to validate the effectiveness of the solutions, involving 100+ hours of data analysis. Proof		
RESEARCH EXPERIENCE			
CFD of fixed bed reactors B Tech Project Jul '25 - Present	Guide: Himanshu Goyal <i>Simulated and optimized reactors using CFD and ML</i> - Simulated 20+ realistic fixed bed reactor configurations using PRCFD to analyze flow, heat, and reaction behaviour. - Benchmarked porous media model results against PRCFD for 4 particles, achieving <10% deviation in key outputs. - Processed 200+ simulation datasets to train an ML model that predicts particle geometry from 5+ target features. Proof		
ChemGPT Research Project Aug '25 - Present	Guide: Niket Kaisare <i>Automated AI system solving graduate-level chemical engineering problems</i> - Built scalable ontology extraction system processing thermo textbooks with 88% precision across 5 knowledge types - Engineered an agentic GraphRAG system achieving a 92% retrieval accuracy utilising iterative hypergraph reasoning. - Demonstrated 94% real-world accuracy across 30+ thermo topics, sustaining transparent step-by-step reasoning. Proof		
Gen AI Content Validation Engine Research Project Jul '25 - Present	Guide: Sudip Das <i>Automated and optimized AI content creation workflows with testing</i> - Authoring a paper quantifying the performance of AI generated content in real-world metrics across 3 channels. - Automated an end-to-end pipeline capable of producing 15 videos daily using LLMs, TTS & dynamic video assembly. - Engineered a multi-agent system to test 5+ unique content strategies, reducing content production time by over 90%. Proof		
Atomic Interactions Research Project May '24 - Aug '24	Guide: Tarak Patra <i>Built deep learning model for accurate atomic energy prediction</i> - Developed and trained a DNN with 3+ layers model to predict energy of atomic interactions with 85% accuracy. - Implemented pre-processing pipelines to extract relevant features from 134,000+ points of the input database. - Utilised various concepts such as One Hot Encoding, Morgan Fingerprinting, and QRChem to implement learning. Proof		
PROJECT WORK			

Blood Flow in Arteries <i>CH2012 Project</i>		- Conducted a very comprehensive review of 10+ articles on blood flow dynamics to inform project methodology. - Utilised 10+ concepts such as Navier-Stokes equations to simulate and analyse flow patterns in arterial networks. - Generated detailed reports of 4 arterial flow simulations, aiding the interpretation and writing of project findings. Proof
Smart Materials <i>CH2014 Project</i>		- Simulated a unique perspiration-based cooling system for high-performance chips to control system temperature. - Developed a codebase that integrated dynamic material properties for conducting finite element thermal analysis. - Analysed 5+ relevant concepts and independently ideated on over 3 improvements and 4 critiques to the paper. Proof
Project Member <i>InstiSocial (XRIG)</i>		- Developed an augmented reality app using Unity as a part of a 3 membered team under Professor Manivannan. - Built an AR-based application from the ground up that guides users to the locations of the 10+ live institute events. - Identified 3 bugs and facilitated the implementation of solutions while reducing reports of problems by over 10%.
Sign Detection <i>Self Project</i>		- Designed a sign language detection system, achieving an accuracy rate of 90% across a dataset of 10,000 images. - Implemented a CNN to recognize and classify over 20 different sign language gestures, processing over 900 fps. - Developed an intuitive interface for real-time sign language detection for more than 5 users during the testing. Proof
Hydrogen Fueled Aircraft <i>Matsci Project</i>		- Designed a CFRP-based cryogenic hydrogen storage tank of 83% gravimetric efficiency, exceeding industry standards - Planned tank dimensions to safely store 5,000 kg of hydrogen for flights of upto 10,000 km carrying 200 passengers. - Developed insulation limiting fuel loss to less than 0.1% per day, combining aerogel, MLI and vacuum insulation. Proof
Modelling a Non Isothermal PFR <i>MLRE Project</i>		- Modeled non isothermal PFR for CO₂ methanation with dispersion, pressure drop, and energy balance equations. - Processed over 5000 data points with StandardScaler, PCA & DBSCAN removing 250 outliers improving data quality. - Designed a neural network, 2 hidden layers, ReLU + sigmoid, achieving an R² score of 85% for conversion prediction. Proof
LEADERSHIP RESPONSIBILITIES		
Inter IIT	Tech '24 <i>Dream 11</i>	- Architected a predictive pipeline with data from 15k+ cricket matches, yielding an MAE < 10 via 500+ Optuna trials - Devised 300+ features with high target correlation (r>0.9), improving the final model's predictive accuracy by 20%+
	Culturals '23	- Selected as a part of the 12-member team under Improv category to participate in Inter-IIT Culturals' Meet 2024. - Shortlisted as a part of the 16-member team to represent the contingent in improv comedy at IITB's MoodIndigo².
Teaching Assistant <i>ID2090</i>		- Supported the course instruction by aiding the professor in making 6+ assignments, doubt solving and discussions. - Provided personalised academic support to 10 students through clarifying core concepts over 20+ doubt sessions. Proof
Social Secretary <i>Narmada Hostel</i>		- Elected by an electorate of 450+ students to develop the culture of the hostel and oversee Hostel Social events. - Responsible for managing a budget of ₹2,00,000, catering to conducting events for uplifting of the Hostel students. Proof
Super Coordinator <i>SR (Saarang)</i>		- Responsible for execution of 6 Milans to be held in 3 cities across the country and 6 imprints in schools in Chennai. - Tasked with ideation of 3+ publicity stunts across Chennai like flash mobs to increase public visibility & awareness.
Coordinator <i>MMCC¹</i>		- Solely responsible for 2 messes throughout the tenure, ensuring their cleanliness, quality and smooth functioning. - Ensured that my allotted messes were ranked 1st amongst 11 messes throughout the full duration of the tenure.
Coordinator <i>Comedy Club</i>		- Single-handedly organised 2 open mic events for stand up comedy and improv competitions during Saarang³ 2024. - Connected with 20+ colleges across the country to invite them to 15+ competitions on campus during the tenure.
EXTRA-CURRICULARS		
Speaking Arts		- Won 2nd prize in "Mad-Ads," where participants had to design an ad for a random product in less than 5 minutes. Proof Emceed for Chemical Engineers Nite in the year 2023-24 in the presence of students of all batches and professors. Proof - Played 3+ lead characters like Birbal and Kabir Das over 10+ years in school plays in both English as well as Hindi.
Distinctions		- Awarded the Certificate of Distinction by Royal Australian Chemical Institute for excelling in their chemistry quiz. Proof - Awarded a Merit Certificate for RMO in class 10 and selected to attend a 3-day camp at Indian Statistical Institute. Proof - Carnatic musician certified at Junior level with over 12 years of experience with 100% marks in the theory paper. Proof
Accolades		- Elected as Class Representative of Chemical Engineering by a class of 120 students against 3 other candidates. Proof - One among the top 3 in school for 10 consecutive years (Class 1-10) to be awarded a prize for General Proficiency. Proof - One of the 60+ students selected for the Infosys "Catch them Young" program for software training in class 8. Proof